

Chapter 45: Women Empowerment through Sericulture: A Socio-economic Perspective with Special Reference to Assam

Subhadeep Seal^{1*} & Sayantika Dey^{1*}

Department of Zoology, Cotton University, Assam, India

*Corresponding E-mail: subhadeepseal20@gmail.com, sayadey30@gmail.com

Abstract:

Women's socio-economic empowerment is essential for inclusive rural development, yet disparities in access to productive resources, credit, technology, education, and decision-making continue to constrain their economic participation. Sericulture has emerged as a women-centric rural enterprise that provides year-round employment, regular income, and entrepreneurship opportunities through a household-based production system. Women constitute nearly 60–70% of the sericulture workforce in the country, participating across the entire value chain including host plant cultivation, silkworm rearing, cocoon production, silk processing, weaving, value addition, and marketing, while in Assam, they play a pivotal role in sustaining Eri and Muga silk production and preserving indigenous knowledge and cultural heritage. Despite the recognised economic importance of sericulture, its contribution to women's empowerment has received comparatively limited comprehensive attention, particularly within the socio-cultural context of Assam. This study critically examines the role of sericulture in enhancing women's income, livelihood diversification, financial inclusion, entrepreneurship, skill development, and participation in household and community decision-making. It also evaluates institutional support mechanisms, government initiatives, and the major challenges affecting women in the sector. The review concludes that strengthening policy support, institutional collaboration, market access, capacity building, and climate-resilient technologies can further enhance the contribution of sericulture to women's empowerment, sustainable livelihoods, and inclusive rural development in Assam.

Keywords: Women empowerment; Sericulture; Socio-economic development; Assam; Sustainable livelihoods; Gender equality; Eri silk; Muga silk.

1. Introduction

Women's empowerment is a fundamental driver of sustainable development, social equity, and inclusive economic growth, as greater participation in economic activities enhances household welfare, reduces poverty, and promotes national development (Kabeer, 1999; United Nations [UN], 2015). Despite their substantial contribution to agriculture, rural women continue to face disparities in access to productive resources, education, financial services, technology, land ownership, and decision-making, limiting their economic independence and livelihood security (UN Women, 2023). In India, promoting women-centred livelihood opportunities has therefore become a key strategy for inclusive rural development. Among allied agricultural sectors, sericulture has emerged as a labour-intensive, environmentally sustainable enterprise that provides year-round employment with relatively low capital investment. Women's participation across the value chain, from host plant cultivation and silkworm rearing to silk processing, weaving, and marketing, strengthens household income, entrepreneurship, and financial inclusion (Central Silk Board [CSB], 2024; Geetha, 2014).

India is the only country producing all four commercially important natural silks: Mulberry, Tasar, Eri, and Muga, while Assam is the exclusive producer of Muga silk and a leading producer of Eri silk, where sericulture is deeply embedded in the livelihoods and cultural heritage of Assamese and indigenous communities (Kalita, 2022; Deuri, 2022). The sector contributes to multiple Sustainable Development

Goals, particularly poverty reduction, gender equality, decent work, and sustainable production (UN, 2015). Although previous studies have primarily focused on production, technology, and marketing, comparatively limited attention has been given to sericulture's multidimensional role in advancing women's socio-economic empowerment in Assam. Accordingly, this study examines women's participation across the sericulture value chain, its contribution to livelihoods, entrepreneurship, financial inclusion, and decision-making, and the institutional support, opportunities, and challenges influencing women's empowerment, with special reference to the Bodoland Territorial Region (BTR).

2. Sericulture: An Overview with Special Reference to India and Assam

Sericulture is an integrated agro-based enterprise encompassing host plant cultivation, silkworm rearing, cocoon production, silk processing, weaving, value addition, and marketing. By integrating agriculture with cottage and textile industries, it generates employment across farm and non-farm sectors with relatively low capital investment, making it well suited to rural and small-holder economies. It also promotes livelihood diversification and environmentally sustainable production through the use of renewable biological resources (FAO, 2022). Originating in ancient China, sericulture spread to India through the Silk Road and became an integral part of the country's rural economy and cultural heritage (Datta & Nanavaty, 2005). India is the only nation producing all four commercially important natural silks: Mulberry (*Bombyx mori*), Tasar (*Antheraea mylitta*), Eri (*Samia ricini*), and Muga (*Antheraea assamensis*), reflecting its rich biodiversity and diverse agro-climatic conditions (CSB, 2024; Ministry of Textiles, 2023). Eri silk is produced without killing the pupa, whereas Muga silk, endemic to Assam, is renowned for its natural golden lustre, durability, and Geographical Indication (GI) status (Choudhury, 2019; Government of India, 2007). Assam is the exclusive producer of Muga silk and one of India's leading producers of Eri silk. Its favourable agro-climatic conditions, abundant host plant resources, and indigenous knowledge have sustained silk cultivation for generations among communities such as the Bodo, Assamese, Mising, Karbi, Rabha, Tiwa, and Deori, where weaving remains closely linked to livelihoods and cultural identity (Borah, 2018; Deuri, 2022). Supported by government initiatives and increasing global demand for sustainable natural fibres, the sector continues to strengthen rural entrepreneurship, women's employment, and inclusive economic growth (Shravanilakshmi *et al.*, 2025).

3. Women's Participation Across the Sericulture Value Chain

Women constitute nearly 60–70% of India's sericulture workforce, making the sector one of the most women-intensive agro-based industries in the country (CSB, 2024). Unlike many rural occupations where women's labour often remains unpaid or undervalued, sericulture provides year-round employment and income through a household-based production system requiring relatively low capital investment. In Assam, women play a pivotal role in sustaining the Eri, Muga, and Mulberry silk industries while preserving indigenous knowledge, weaving traditions, and the state's rich cultural heritage (Geetha, 2014; Borah, 2018). Their responsibilities span the entire production cycle, beginning with host plant cultivation, including nursery management, irrigation, pruning, nutrient management, pest control, and harvesting of castor (*Ricinus communis*), kesseru (*Heteropanax fragrans*), som (*Persea bombycina*), and soalu (*Litsea monopetala*), which are essential for healthy silkworm growth and quality cocoon production (Government of Assam, 2023).

Women also dominate the labour-intensive silkworm rearing stage, undertaking egg incubation, feeding, hygienic rearing, environmental regulation, and disease management to ensure high cocoon yield and silk

quality (Datta & Nanavaty, 2005). They are further involved in cocoon harvesting, grading, drying, storage, and post-cocoon operations such as reeling, spinning, twisting, degumming, yarn preparation, and weaving. Traditional tools such as the Takuri and Charkha, together with improved technologies, enable the production of value-added products including mekhela chadors, sarees, shawls, stoles, scarves, furnishings, and handicrafts (CSB, 2024; Government of Assam, 2023).

Beyond production, women increasingly participate in entrepreneurship and marketing through Self-Help Groups (SHGs), cooperatives, Farmer Producer Organisations (FPOs), and microenterprises engaged in branding, retailing, exhibitions, and digital commerce. Government support through technical training, institutional credit, and market linkages has strengthened enterprise development and income generation. However, limited digital literacy, inadequate marketing infrastructure, and market fluctuations continue to constrain the growth and competitiveness of women-led enterprises, underscoring the need for sustained institutional and market support (CSB, 2024; Geetha, 2014).

Table 1. Participation of Different Age Groups of Women in Sericulture Activities

Source: Adapted from Madhu *et al.* (2023).

Age Group	Participation (%)	Key Activities	Remarks
15–25 years	24	Silkworm rearing, cocoon harvesting, leaf harvesting	Young women actively participate but at a lower level than the middle-aged group.
26–45 years	60	Silkworm feeding, cocoon harvesting, mounting, leaf harvesting	Most active and economically productive age group in sericulture.
46 years and above	16	Limited participation	Participation declines due to reduced physical involvement in labour-intensive activities.

Table 2. Share of Women in Different Activities of Sericulture

Source: Adapted from Madhu *et al.* (2023).

Sl. No.	Activities	Women (%)	Men (%)	Total (%)
1	Establishment of mulberry plantation	51.38	48.62	100.00
2	Mulberry maintenance	57.00	43.00	100.00
3	Silk cocoon production	60.02	39.98	100.00

4	Raw silk production	62.30	37.70	100.00
---	---------------------	-------	-------	--------

4. Socio-economic Dimensions of Women's Empowerment through Sericulture

Sericulture significantly contributes to the socio-economic empowerment of rural women by providing regular employment and income through activities such as host plant cultivation, silkworm rearing, cocoon production, silk processing, weaving, and marketing. Its household-based production system and low capital requirement enable women to combine livelihood activities with domestic responsibilities, thereby improving household income, financial security, expenditure on education, healthcare, and nutrition, and encouraging savings and entrepreneurship. Participation in Self-Help Groups (SHGs), cooperatives, and microfinance programmes further strengthens women's access to institutional credit, financial inclusion, and enterprise development, enhancing their economic independence (CSB, 2024; National Rural Livelihood Mission [NRLM], 2023).

Beyond economic benefits, sericulture enhances women's social status, self-confidence, decision-making capacity, and leadership through skill development, technical training, and participation in community-based institutions. The expansion of the sericulture value chain has created opportunities for women-led enterprises in weaving, natural dyeing, handicrafts, and other value-added silk products, supported by government initiatives, digital marketing, and entrepreneurship programmes. From a sustainable development perspective, sericulture strengthens human, financial, social, natural, and physical capital, contributing to resilient rural livelihoods, biodiversity conservation, and inclusive economic growth. These multidimensional benefits directly support several Sustainable Development Goals, including SDG 1 (No Poverty), SDG 5 (Gender Equality), SDG 8 (Decent Work and Economic Growth), and SDG 12 (Responsible Consumption and Production), highlighting sericulture as an effective pathway for women's empowerment and sustainable rural development (UN, 2015).

5. Institutional Support and Government Initiatives for Women in Sericulture

Institutional support has played a pivotal role in enhancing women's participation in sericulture by improving access to training, technology, finance, infrastructure, and markets. In India, the development of the sericulture sector is primarily coordinated by the Central Silk Board (CSB) under the Ministry of Textiles, in collaboration with state agencies such as the Directorate of Sericulture, Government of Assam. Through initiatives including the Beneficiary Empowerment Programme (BEP), Skill Training and Enterprise Development Programme (STEP), Sericulture Resource Centres (SRCs), and the flagship Silk Samagra-2 scheme, these institutions promote quality silkworm seed production, host plant cultivation, post-cocoon infrastructure, technology dissemination, entrepreneurship development, and market linkages. In Assam, these interventions have been particularly important for strengthening traditional Eri and Muga sericulture by providing technical guidance, capacity-building programmes, extension services, and support for weaving, natural dyeing, product diversification, and value-added silk enterprises, thereby improving the livelihoods and income-generating opportunities of rural women (CSB, 2024; Ministry of Textiles, 2023; Government of Assam, 2023).

Community-based institutions and financial support mechanisms have further strengthened women's empowerment in sericulture. Self-Help Groups (SHGs), supported by the National Rural Livelihood

Mission (NRLM) and the Assam State Rural Livelihoods Mission (ASRLM), have facilitated access to savings, credit, skill development, collective marketing, and enterprise management, while financial assistance from the National Bank for Agriculture and Rural Development (NABARD), cooperative institutions, and commercial banks has encouraged the establishment of women-led sericulture enterprises. Nevertheless, challenges such as limited financial literacy, inadequate market access, insufficient value-addition facilities, uneven extension services, climate variability, and fluctuating silk prices continue to constrain the sector's full potential. Addressing these issues through strengthened digital extension services, climate-resilient production technologies, entrepreneurship promotion, gender-responsive financial policies, and improved value-chain integration will be essential for enhancing the long-term sustainability and socio-economic impact of women-led sericulture in rural India (NRLM, 2023; FAO, 2022; NABARD, 2023).

6. Case Study: Women-led Sericulture in the Bodoland Territorial Region (BTR), Assam

The Bodoland Territorial Region (BTR), comprising Kokrajhar, Chirang, Baksa, Tamulpur, and Udalguri districts, is one of Assam's principal sericulture-producing regions, particularly for Eri silk (CSB, 2024; Directorate of Sericulture, Assam, 2023). Traditionally practised by indigenous communities, especially the Bodo, sericulture is closely intertwined with household livelihoods and cultural heritage. Its low investment requirement and year-round employment potential have made it a sustainable livelihood option, with women playing a pivotal role throughout the value chain (Borah, 2018; Geetha, 2014). Women actively participate in host plant cultivation, silkworm rearing, cocoon collection, spinning, weaving, and the production of value-added Eri silk products (Choudhury, 2019). The household-based nature of the enterprise enables them to combine domestic responsibilities with income-generating activities, thereby improving household expenditure on education, healthcare, nutrition, and savings while enhancing their participation in household decision-making and local economic activities (Borah, 2018). Institutional support from the Central Silk Board, the Directorate of Sericulture, Government of Assam, Silk Samagra, and the NRLM has strengthened the sector through quality silkworm seed, technical training, extension services, entrepreneurship development, and market linkages (CSB, 2024; Ministry of Textiles, 2023).

Several targeted initiatives have further accelerated the growth of women-led sericulture in the region. Project *Thaokhri*, implemented by the Bodoland Territorial Council (BTC), promoted scientific silkworm rearing, host plant expansion, improved spinning technologies, and value addition, while the SILKS Portal developed by the North Eastern Space Applications Centre (NESAC) supports GIS- and remote sensing-based planning for host plant cultivation and extension services (BTC, 2021). These interventions have enhanced productivity, strengthened entrepreneurship, and improved livelihood opportunities for rural women.

The BTR exemplifies how traditional knowledge, institutional support, and community participation can collectively strengthen women's socio-economic empowerment through sericulture. Although increased income, financial inclusion, leadership through SHGs, and diversification into weaving and value-added products have improved rural livelihoods, challenges including climate variability, pest incidence, inadequate post-cocoon infrastructure, weak market linkages, and limited access to finance and digital marketing continue to constrain enterprise growth. Strengthening climate-smart production, post-cocoon modernisation, branding, digital marketing, and women-led producer enterprises will be essential for

ensuring the long-term sustainability and competitiveness of the region's sericulture sector (CSB, 2024; FAO, 2022; Ministry of Textiles, 2023).

7. Challenges and Constraints to Women's Empowerment through Sericulture

Despite its significant contribution to rural livelihoods, the potential of sericulture to enhance women's socio-economic empowerment remains constrained by financial, institutional, technological, and market-related barriers. Women, who constitute a major share of the sericulture workforce, often face limited access to institutional credit, productive resources, modern technologies, quality inputs, and extension services due to inadequate collateral, low financial literacy, and procedural complexities. In addition, weak market linkages, insufficient post-cocoon infrastructure, dependence on intermediaries, and limited access to digital marketing platforms reduce productivity, value addition, and income from women-led sericulture enterprises (CSB, 2024; NABARD, 2023).

Climate variability and persistent socio-cultural inequalities further restrict the sustainable development of women-led sericulture. Rising temperatures, erratic rainfall, floods, and increased pest and disease incidence adversely affect host plant cultivation, silkworm rearing, and cocoon production, while household responsibilities, limited asset ownership, restricted mobility, and inadequate representation in decision-making continue to hinder women's entrepreneurial growth and leadership. Addressing these challenges requires an integrated approach involving climate-resilient production technologies, improved market access, strengthened extension services, gender-responsive financial support, and policies that ensure equitable access to resources, training, and leadership opportunities for women in the sericulture sector (FAO, 2022; Kabeer, 1999).

8. Future Prospects and Policy Recommendations

The growing global demand for natural, sustainable, and ethically produced textiles presents substantial opportunities for expanding India's sericulture sector, particularly the Eri and Muga silk industries of Assam. Future growth will depend on strengthening women-led enterprises through improved access to skill development, modern technologies, financial services, digital advisory platforms, climate-smart production practices, and value-added silk products. Greater institutional collaboration among government agencies, research organisations, financial institutions, SHGs, and producer organisations will enhance women's entrepreneurship, market access, income generation, and livelihood resilience while supporting sustainable rural development (CSB, 2024; Ministry of Textiles, 2023).

To maximise the contribution of sericulture to women's socio-economic empowerment, policy interventions should prioritise gender-responsive access to credit, insurance, technical training, entrepreneurship development, climate-resilient technologies, and digital marketing, alongside strengthening value-chain integration, branding, and market infrastructure. Special emphasis should be placed on promoting women-led producer enterprises in the BTR region, where Eri silk production offers significant opportunities for tribal livelihoods and rural entrepreneurship. These interventions will further strengthen the sector's contribution to the Sustainable Development Goals by reducing poverty, advancing gender equality, promoting decent work and economic growth, encouraging responsible production, and supporting biodiversity conservation (FAO, 2022; UN, 2015).

9. Conclusion

This study highlights sericulture as a transformative rural enterprise that promotes women's socio-economic empowerment in Assam. By engaging women across the entire value chain, that is, from host plant cultivation and silkworm rearing to silk processing, weaving, value addition, entrepreneurship, and marketing, it enhances household income, financial inclusion, leadership, livelihood resilience, and preserves the state's Eri and Muga silk heritage. These multidimensional benefits support inclusive rural development and contribute to poverty reduction, gender equality, decent work, responsible production, and biodiversity conservation. However, the sector's full potential remains constrained by limited access to finance, technology, infrastructure, markets, institutional support, and climate-related challenges. Addressing these barriers requires gender-responsive policies that strengthen skill development, digital and financial inclusion, climate-resilient technologies, value-chain integration, entrepreneurship, and institutional collaboration, with particular emphasis on women-led enterprises in the Bodoland Territorial Region. With sustained policy support, research-driven innovation, and coordinated stakeholder efforts, sericulture can continue to drive women's empowerment, inclusive economic growth, and sustainable rural transformation.

References

- [1]. Bodoland Territorial Council. (2021). *Project Thaokhri: Annual progress report*. Bodoland Territorial Council, Kokrajhar, Assam.
- [2]. Borah, M. (2018). *Traditional sericulture and livelihood among the Bodo community of Assam*. Department of Sericulture, Government of Assam.
- [3]. Central Silk Board. (2024). Ministry of Textiles, Government of India.
- [4]. Choudhury, S. (2019). *Eri silk culture and rural livelihood development in Assam*. Directorate of Sericulture, Government of Assam.
- [5]. Datta, R. K., & Nanavaty, M. (2005). *Global silk industry: A complete source book*.
- [6]. Deuri, P. (2022). Traditional weaving and indigenous silk heritage of Assam. *Journal of Rural and Agricultural Research*, 22(2), 54–63.
- [7]. Food and Agriculture Organisation of the United Nations. (2022). *The future of sustainable agriculture and natural fibres*.
- [8]. Geetha, G. S. (2014). Women empowerment through sericulture in India. *Indian Silk*, 53(6), 26–31.
- [9]. Government of India. (2007). *Geographical indication registration of Muga silk*. Geographical Indications Registry.
- [10]. Government of Assam. (2023). *Annual administrative report 2022–2023*. Directorate of Sericulture, Government of Assam.
- [11]. Kabeer, N. (1999). Resources, agency, achievements: Reflections on the measurement of women's empowerment. *Development and Change*, 30(3), 435–464.
- [12]. Kalita, T. (2022). *Sericulture: With special reference to Assam*. EBH Publishers.
- [13]. Madhu, M., Ramesh, B., Kumar, S., & Devi, P. (2023). Women's participation in sericulture activities: An empirical study. *Indian Journal of Extension Education*, 59(3), 142–147.
- [14]. Ministry of Textiles. (2023). *Annual report 2022–2023*. Government of India.
- [15]. National Bank for Agriculture and Rural Development. (2023). *Annual report 2022–2023*. NABARD.

-
- [16]. National Rural Livelihoods Mission. (2023). *Annual report 2022–2023*. Ministry of Rural Development, Government of India.
- [17]. North Eastern Space Applications Centre. (2023). *SILKS Portal: GIS-based decision support system for sericulture*.
- [18]. Shravanilakshmi, R., Kumar, V., Singh, P., & Das, A. (2025). Sustainable silk production and rural entrepreneurship in India. *Journal of Natural Fibres*.
- [19]. UN Women. (2023). *Progress on the Sustainable Development Goals: The gender snapshot 2023*. UN Women.
- [20]. United Nations. (2015). *Transforming our world: The 2030 Agenda for Sustainable Development*.